# LIFTING RANK-TWO CHOI BOUNDS TO $r$-POSITIVE MAPS

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Abstract. Let $\Phi : M_d(\mathbb{C}) \to M_d(\mathbb{C})$ be a completely positive map with a transpose-symmetric Kraus representation. Its Choi operator determines the rank-two constant

$$\beta_2(\Phi) = \frac{1}{2} \sup_{\substack{\|z\|=1 \\ \mathrm{SR}(z)\leq 2}} \langle z, J(\Phi)z\rangle.$$

We prove that, for every $r \geq 1$,

$$\Phi \circ \tau + (r-1)\beta_2(\Phi)\left(\Delta - \frac{1}{r}\mathrm{id}\right)$$

is $r$-positive, where $\tau$ is transposition and $\Delta(X) = \mathrm{Tr}(X)I$. The proof turns the negative spectrum of a partially transposed rank-one operator into the edges of the complete graph $K_r$. Regrouping by vertices costs its degree $r-1$, rather than its $\binom{r}{2}$ edges. Transpose symmetry protects one direction and is exactly what produces the term $-\mathrm{id}/r$.

For $\Phi = \Lambda_U \otimes \Lambda_V$, with $\Lambda_E(X) = \mathrm{Tr}(X)I - X^T$, a sharp double-skew singular-value estimate gives $\beta_2 \leq 1$, with equality when both local dimensions are at least two. A map identity then yields, for every operator $C$ of rank at most $r$ on $U \otimes V$,

$$\|\mathrm{Tr}_U C\|_2^2 + \|\mathrm{Tr}_V C\|_2^2 \leq r\|C\|_2^2 + \frac{1}{r}|\mathrm{Tr}\, C|^2,$$

with dimension-uniform coefficients that are sharp whenever one local dimension is at least $r$ and the other is at least two. A second family, built from the edges of an arbitrary graph, has the same local constant and attains the universal coefficient $r-1$ whenever the graph contains an $r$-clique. The Choi lifting chain through the partial-trace and graph results, together with its sharpness witnesses, is formalized in Lean 4.

**AI-generation and verification disclosure.** The author selected the problem, supplied the prompts, and directed the iterations. The arguments, applications, and exposition were generated with OpenAI GPT-5.6 and Anthropic Claude Opus 5; the Lean development was generated with Claude Fable 5 and Claude Opus 5. The author checked the formal–informal interfaces but did not independently derive every proof. Lean checks deductions, not whether the formal definitions faithfully represent the informal mathematics. The manuscript has not received independent expert review.

## 1. The result behind the motivating inequality

The motivating statement is a rank-sensitive relation among the two marginals of an arbitrary bipartite matrix. If $C \in \mathcal{L}(U \otimes V)$ and $\operatorname{rank} C \leq r$, then

$$\|\mathrm{Tr}_U C\|_2^2 + \|\mathrm{Tr}_V C\|_2^2 \leq r\|C\|_2^2 + \frac{1}{r}|\mathrm{Tr}\, C|^2. \tag{1}$$

Here $\|\cdot\|_2$ is the Hilbert–Schmidt norm and $\mathrm{Tr}_U$ means that the $U$ factor is traced out. Neither normality nor self-adjointness is assumed. The coefficients are sharp when one local dimension is at least $r$ and the other is at least two.

Costa Rico proved Equation (1) for normal matrices and connected its rank-two case with the two-copy distillability problem for Werner states [3]; Costa Rico and Wolf identified its extension beyond normal matrices as an open problem [4]. Fu, Gao, and Park proved the sharp rank-two

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double-skew estimate that supplies the local input used here [7]. The arbitrary-$r$ inequality also appears, by a balanced-factorization route, in Fraser, Huber, Pozsgay, and Vona [6]; the two lifting mechanisms are compared in Section 5.1. Song and Chen [12] and Bharti, Gajjala, and Haug [1] prove the rank-two case and further consequences. The contributions of this account are the general rank-two-to-$r$ lifting theorem, the protected direction supplied by transpose symmetry, the graph family showing universal sharpness, and the formal verification. The partial-trace inequality is the flagship application and a point of overlap with the cited work.

Fix orthonormal bases whenever a transpose is used. For a completely positive map

$$\Phi(X) = \sum_a A_a X A_a^*$$

on $\mathcal{L}(H)$, let

$$J(\Phi) = \sum_a |\operatorname{vec} A_a\rangle\langle \operatorname{vec} A_a|$$

be its Choi operator. For a positive operator $X$ on a tensor product, set

$$\|X\|_{S(k)} := \sup_{\substack{\|z\|=1 \\ \mathrm{SR}(z)\le k}} \langle z, Xz\rangle.$$

The convention $\operatorname{vec}(|x\rangle\langle y|) = x \otimes \overline{y}$ gives $\mathrm{SR}(\operatorname{vec} C) = \operatorname{rank} C$.

The *Kraus space* of $\Phi$ is

$$\mathcal{K}_\Phi := \operatorname{span}\{A_a\};$$

equivalently, its vectorization is the range of $J(\Phi)$, so it is independent of the Kraus representation. We say that it has *even transpose parity* when $K^T = K$ for every $K \in \mathcal{K}_\Phi$.

The rank-two Choi constant. For a completely positive map $\Phi$, define

$$\beta_2(\Phi) := \frac{1}{2}\|J(\Phi)\|_{S(2)}. \tag{2}$$

This depends on the map, not on a chosen Kraus representation. If $K_c = \sum_a c_a A_a$, then the projection duality for $S(k)$-norms [9] gives the useful equivalent formula

$$2\beta_2(\Phi) = \sup_{\|c\|_2=1} \left(s_1(K_c)^2 + s_2(K_c)^2\right), \tag{3}$$

where $\|c\|_2$ is the Euclidean coefficient norm and the singular values are in decreasing order, padded by zeros beyond the matrix dimension.

**Theorem 1.1** (Rank-two lifting)**.** *Let $\Phi : \mathcal{L}(H) \to \mathcal{L}(H)$ be completely positive. For every integer $r \ge 1$, the map*

$$\Phi \circ \tau + (r-1)\beta_2(\Phi)\Delta \tag{4}$$

*is $r$-positive.*

*If $\mathcal{K}_\Phi$ has even transpose parity, then the sharper map*

$$\Psi_{r,\Phi} := \Phi \circ \tau + (r-1)\beta_2(\Phi)\left(\Delta - \frac{1}{r}\mathrm{id}\right), \qquad \Delta(X) = \mathrm{Tr}(X)I, \tag{5}$$

*is $r$-positive.*

Here a map is $r$-positive when its ampliation by the identity on $M_r(\mathbb{C})$ is positive. Equivalently, its Choi operator has nonnegative expectation on every vector of Schmidt rank at most $r$ [2, 9].

**Corollary 1.2** (Using an upper bound for the local constant)**.** *Under the protected hypothesis of Theorem 1.1, let $r \ge 1$ be an integer and let $\gamma \ge \beta_2(\Phi)$. Then*

$$\Phi \circ \tau + (r-1)\gamma\left(\Delta - \frac{1}{r}\mathrm{id}\right)$$

*is $r$-positive.*

*Proof.* The map $\Delta - \frac{1}{r}\mathrm{id}$ is $r$-positive: for every $C$ with $\operatorname{rank} C \le r$, its Choi quadratic form is

$$\|C\|_2^2 - \frac{1}{r}|\operatorname{Tr} C|^2 \ge 0$$

by the trace–rank inequality. Add $(r-1)(\gamma - \beta_2(\Phi))(\Delta - \frac{1}{r}\mathrm{id})$ to the protected map of Theorem 1.1. □

The theorem is not an induction on $r$, nor does it tensorize a rank-two statement. A partial transpose produces pair-indexed negative modes at level $r$; the rank-two Choi bound controls each mode, and a complete-graph estimate assembles them. This distinction is useful when looking for new instances: one needs a local two-vector estimate, not a recursive relation between successive levels.

## 2. Using the theorem

Figure 1 separates the universal argument from the work specific to an application. The first row gives the unprotected theorem; the second adds even transpose parity and the protected direction. The final application-specific identity is optional: it is needed only to translate $r$-positivity into a concrete inequality or inclusion.

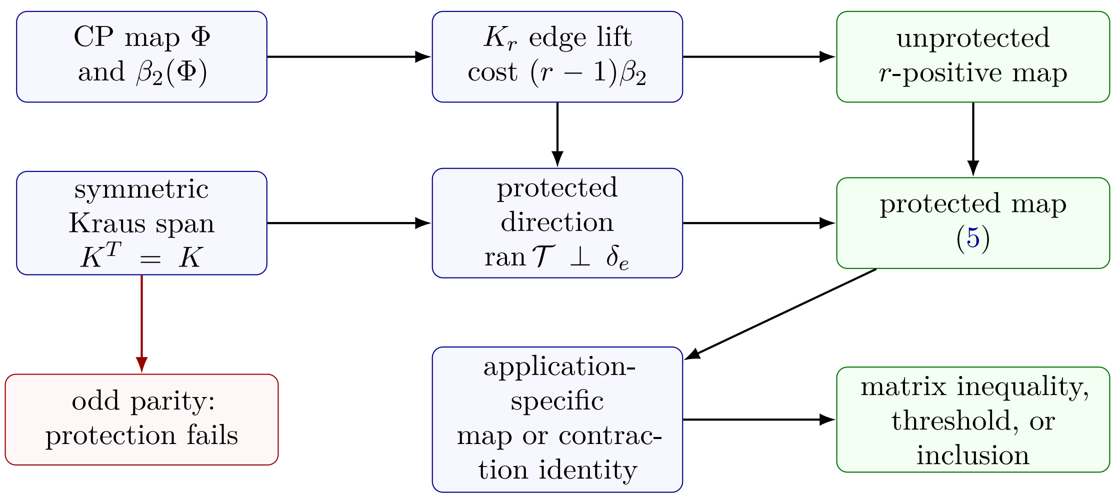


FIGURE 1. A diagnostic for applying the rank-two lifting theorem. The rank-two constant and even transpose parity are universal inputs. Concrete downstream formulae require a separate identity for the chosen map.

For a prospective application, four questions suffice.

(1) Is $\Phi$ completely positive, and does its Kraus span have even transpose parity? Complete positivity gives Equation (4); even parity gives the protected pencil Equation (5).
(2) Can one bound the two-vector quantity $s_1(K_c)^2 + s_2(K_c)^2$? This determines the universal sufficient coefficient $(r-1)\beta_2(\Phi)$.
(3) Is that coefficient necessary for the particular map? This requires a rank-$r$ Choi witness; saturation of the local two-vector bound alone need not propagate through the complete-graph estimate.
(4) Is there a map or contraction identity that turns the corrected Choi form into the desired matrix expression? Without one, the conclusion is $r$-positivity, not the proposed matrix inequality.

The improvement from Equation (4) to Equation (5) is a rank-one protection, not a change in the complete-graph estimate. The example in Section 7 shows that this improvement genuinely fails for an odd Kraus space.

## 3. The complete-graph proof

The proof is included in the main text because its two structural steps—the edge decomposition and the protected direction—are the information needed to adapt the theorem.

Let $e_1, \dots, e_s$ be an orthonormal family in $H$, let $q_1, \dots, q_s$ be the standard basis of $Q = \mathbb{C}^s$, and set

$$\delta_e = \sum_{i=1}^{s} e_i \otimes q_i, \qquad \rho_e = |\delta_e\rangle\langle\delta_e|, \qquad \Pi_e = \frac{1}{s}\rho_e.$$

The partial transpose of $\rho_e$ on $H$ is the flip between $\operatorname{span}\{\overline{e}_i\}$ and $Q$. Its negative spectral projection is

$$P_- = \sum_{1 \le i < j \le s} |\eta_{ij}\rangle\langle\eta_{ij}|, \qquad \eta_{ij} = \frac{\overline{e}_i \otimes q_j - \overline{e}_j \otimes q_i}{\sqrt{2}}. \tag{6}$$

Writing $P_+$ for the positive projection,

$$\rho_e^{T_H} = P_+ - P_-. \tag{7}$$

The vectors $\eta_{ij}$ are naturally indexed by the edges of $K_s$.

**Proposition 3.1** (Negative-sector estimate)**.** *For every completely positive $\Phi$ and every orthonormal family as above,*

$$(\Phi \otimes \mathrm{id}_Q)(P_-) \preceq (s-1)\beta_2(\Phi) I. \tag{8}$$

*If $\Phi$ has a transpose-symmetric Kraus representation, then*

$$(\Phi \otimes \mathrm{id}_Q)(P_-) \preceq (s-1)\beta_2(\Phi)(I - \Pi_e). \tag{9}$$

*Proof.* Choose Kraus operators $\{A_a\}$ for $\Phi$. For coefficients $c = (c_a^{(ij)})$, indexed by edges $i < j$ and Kraus labels $a$, define

$$\mathcal{T}c = \sum_{i<j} \sum_{a} c_a^{(ij)} (A_a \otimes I_Q)\eta_{ij}, \qquad K^{(ij)} = \sum_a c_a^{(ij)} A_a.$$

The Kraus formula and Equation (6) give

$$\mathcal{T}\mathcal{T}^* = (\Phi \otimes \mathrm{id}_Q)(P_-). \tag{10}$$

Grouping the output by vertices rather than edges,

$$\mathcal{T}c = \frac{1}{\sqrt{2}} \sum_{k=1}^{s} \left( \sum_{i<k} K^{(ik)}\overline{e}_i - \sum_{k<j} K^{(kj)}\overline{e}_j \right) \otimes q_k. \tag{11}$$

There are $s-1$ incident terms at each vertex. Cauchy–Schwarz at each vertex therefore gives

$$\|\mathcal{T}c\|^2 \le \frac{s-1}{2} \sum_{i<j} \left( \|K^{(ij)}\overline{e}_i\|^2 + \|K^{(ij)}\overline{e}_j\|^2 \right). \tag{12}$$

For any orthonormal $x, y$ and $K = \sum_a c_a A_a$,

$$\|Kx\|^2 + \|Ky\|^2 \le 2\beta_2(\Phi)\|c\|_2^2. \tag{13}$$

Indeed, if $P = |x\rangle\langle x| + |y\rangle\langle y|$, the Ky Fan variational principle and the homogeneous form of Equation (3) give

$$\|Kx\|^2 + \|Ky\|^2 = \|KP\|_2^2 \le s_1(K)^2 + s_2(K)^2 \le 2\beta_2(\Phi)\|c\|_2^2.$$

Substitution into Equation (12) yields

$$\|\mathcal{T}c\|^2 \le (s-1)\beta_2(\Phi)\|c\|_2^2. \tag{14}$$

Together with Equation (10), this proves Equation (8).

Suppose now that $A_a^T = A_a$. Then every $K^{(ij)}$ is symmetric, and

$$\langle \delta_e, \mathcal{T}c\rangle = \frac{1}{\sqrt{2}}\sum_{i<j}\left(\overline{e}_j^T K^{(ij)}\overline{e}_i - \overline{e}_i^T K^{(ij)}\overline{e}_j\right) = 0.$$

Thus $\operatorname{ran}\mathcal{T} \subseteq \delta_e^\perp$, and therefore

$$\mathcal{T}\mathcal{T}^* = (I-\Pi_e)\mathcal{T}\mathcal{T}^*(I-\Pi_e) \preceq \|\mathcal{T}\|^2(I-\Pi_e).$$

Together with Equation (14), this proves Equation (9). □

*Proof of Theorem 1.1.* By Equation (7), complete positivity of $\Phi$, and Equation (8),

$$(\Phi\otimes \mathrm{id}_Q)(\rho_e^{T_H}) + (s-1)\beta_2(\Phi)I \succeq 0.$$

Here $I = (\Delta\otimes\mathrm{id}_Q)(\rho_e)$ because $\operatorname{Tr}_H \rho_e = I_Q$. Filtering on $Q$ produces every generating vector of Schmidt rank at most $s$, and hence proves positivity on every corresponding rank-one input. Thus the map in Equation (4) is $s$-positive at parameter $s$. For a test vector of Schmidt rank $s \le r$, increasing the parameter adds $(r-s)\beta_2(\Phi)\Delta$, a completely positive map. This proves the unprotected assertion for every $r$.

Suppose now that the Kraus operators are transpose-symmetric. Replacing Equation (8) by Equation (9) gives

$$(\Phi\otimes \mathrm{id}_Q)(\rho_e^{T_H}) + (s-1)\beta_2(\Phi)(I-\Pi_e) \succeq 0. \tag{15}$$

Since $\operatorname{Tr}_H \rho_e = I_Q$ and $\rho_e = s\Pi_e$,

$$I - \Pi_e = \left[\left(\Delta - \frac{1}{s}\mathrm{id}\right)\otimes \mathrm{id}_Q\right](\rho_e).$$

Hence Equation (15) says $(\Psi_{s,\Phi}\otimes\mathrm{id}_Q)(\rho_e)\succeq 0$. Filtering on $Q$ proves the same assertion for every rank-one positive operator whose generating vector has exact Schmidt rank $s$.

It remains to test the Choi operator of $\Psi_{r,\Phi}$ on a vector of Schmidt rank $s\le r$. Let $C$ be its inverse vectorization, so $\operatorname{rank} C = s$. The difference between the $\Psi_{r,\Phi}$ and $\Psi_{s,\Phi}$ Choi quadratic forms is

$$(r-s)\beta_2(\Phi)\left(\|C\|_2^2 - \frac{|\operatorname{Tr} C|^2}{rs}\right) \ge 0, \tag{16}$$

because $|\operatorname{Tr} C|^2 \le s\|C\|_2^2$. Thus the Choi operator of $\Psi_{r,\Phi}$ is nonnegative on every vector of Schmidt rank at most $r$, which is equivalent to $r$-positivity. □

The use of exact Schmidt rank $s$ in the final paragraph matters when $r > \dim H$: an orthonormal $r$-frame in $H$ need not exist. No padding argument is required.

## 4. The double-skew input

For a finite-dimensional coordinate Hilbert space $E$, define

$$\Lambda_E(X) = \operatorname{Tr}(X)I_E - X^T.$$

If $L_{ab} = E_{ab} - E_{ba}$, then

$$\Lambda_E(X) = \sum_{a<b} L_{ab}XL_{ab}^*. \tag{17}$$

Thus $\Lambda_E$ is completely positive. The Kraus operators of $\Lambda_U\otimes\Lambda_V$ are $L_{ab}\otimes L_{cd}$, which are transpose-symmetric because both factors are skew-symmetric.

The value of its Choi constant is the only map-specific input needed by Theorem 1.1.

**Theorem 4.1** (Double-skew action bound)**.** *Let*

$$\mathfrak{so}(E) = \{L \in \mathcal{L}(E) : L^T = -L\}.$$

*For every* $K \in \mathfrak{so}(U) \otimes \mathfrak{so}(V)$ *and every orthonormal* $x, y \in U \otimes V$*,*

$$\|Kx\|^2 + \|Ky\|^2 \leq \frac{1}{2}\|K\|_2^2. \tag{18}$$

This is equivalent to the sharp Schmidt-rank-two projection estimate of Fu, Gao, and Park [7]. The proof in Section A adapts and reorganizes the argument of [7, §2], using Autonne–Takagi directly; the following roadmap isolates its delicate point. The abstract lifting theorem consumes only the resulting local constant and is independent of this concrete proof. Tensor products of two skew matrices are symmetric under full transposition, so Autonne–Takagi writes

$$K = \sum_i s_i w_i w_i^T.$$

Let $\Pi$ be the Hilbert–Schmidt projection onto the double-skew space. For every two-term Takagi truncation $N = s_a w_a w_a^T + s_b w_b w_b^T$, a partial-transpose calculation gives

$$\langle N, N^{T_U}\rangle_{\mathrm{HS}} \geq 0, \qquad \langle N, \Pi N\rangle_{\mathrm{HS}} \leq \frac{1}{2}\|N\|_2^2.$$

Projecting the truncation back against $K$ then forces

$$s_a^2 + s_b^2 \leq \frac{1}{2}\|K\|_2^2 \quad \text{for every pair } a \neq b.$$

A two-coordinate weight argument turns these pair inequalities into Equation (18). The nonnegativity after partial transpose is where the tensor product of the two skew spaces enters.

**Corollary 4.2** (Exact local constant)**.** *For* $\Phi = \Lambda_U \otimes \Lambda_V$*,*

$$\beta_2(\Phi) \leq 1.$$

*Equality holds if* $\dim U, \dim V \geq 2$*. If either dimension is less than two, then* $\Phi = 0$ *and* $\beta_2(\Phi) = 0$*.*

*Proof.* The Kraus products in Equation (17) are pairwise Hilbert–Schmidt orthogonal and have squared norm 4. Thus, for $K_c = \sum c_{ab,cd} L_{ab} \otimes L_{cd}$,

$$\|K_c\|_2^2 = 4\|c\|_2^2.$$

Theorem 4.1 and Equation (3) give $2\beta_2 \leq 2$. If both factors contain two distinct basis vectors, a single Kraus product has four singular values equal to one; its two largest squared singular values sum to 2, proving equality. In a space of dimension below two there are no skew Kraus operators. ☐

## 5. The partial-trace inequality

The general theorem becomes a marginal inequality only because $\Lambda_U \otimes \Lambda_V$ satisfies a second, application-specific identity. Put

$$\mathcal{R}^E_{-a} = \Delta_E - a\,\mathrm{id}_E.$$

Since $\Lambda_E \circ \tau_E = \Delta_E - \mathrm{id}_E$, a direct expansion in the two-dimensional algebra $\mathrm{span}\{\Delta, \mathrm{id}\}$ gives, for every integer $r \geq 1$,

$$(\Lambda_U \otimes \Lambda_V) \circ \tau + (r-1)\left(\Delta_{U\otimes V} - \frac{1}{r}\mathrm{id}\right) = r\,\mathcal{R}^U_{-1/r} \otimes \mathcal{R}^V_{-1/r}. \tag{19}$$

By Theorem 4.2, $\beta_2(\Lambda_U \otimes \Lambda_V) \leq 1$ in every dimension. Applying Theorem 1.2 with $\gamma = 1$ shows that the left side of Equation (19), and hence the map on the right, is $r$-positive. This includes the degenerate cases in which the sharp local constant is zero.

**Theorem 5.1** (Sharp rank-$r$ partial-trace inequality)**.** *Let $r \geq 1$ be an integer, and let $U, V$ be finite-dimensional complex Hilbert spaces. For every $C \in \mathcal{L}(U \otimes V)$ with* $\operatorname{rank} C \leq r$*,*

$$\|\operatorname{Tr}_U C\|_2^2 + \|\operatorname{Tr}_V C\|_2^2 \leq r\|C\|_2^2 + \frac{1}{r}|\operatorname{Tr} C|^2. \tag{20}$$

*If $C \neq 0$ and $s = \operatorname{rank} C$, the stronger exact-rank form*

$$\|\operatorname{Tr}_U C\|_2^2 + \|\operatorname{Tr}_V C\|_2^2 \leq s\|C\|_2^2 + \frac{1}{s}|\operatorname{Tr} C|^2 \tag{21}$$

*holds. Consequently, equality in Equation (20) for nonzero $C$ requires* $\operatorname{rank} C = r$*.*

*Proof.* The Choi quadratic form of the right side of Equation (19), tested on $\operatorname{vec} C$, is

$$r\|C\|_2^2 + \frac{1}{r}|\operatorname{Tr} C|^2 - \|\operatorname{Tr}_U C\|_2^2 - \|\operatorname{Tr}_V C\|_2^2. \tag{22}$$

This follows by expanding the tensor product: the four terms $\Delta \otimes \Delta$, $\Delta \otimes \mathrm{id}$, $\mathrm{id} \otimes \Delta$, and $\mathrm{id} \otimes \mathrm{id}$ contract respectively to $\|C\|_2^2$, the two marginal Hilbert–Schmidt norms, and $|\operatorname{Tr} C|^2$, with the coefficients in Equation (19). Since $r$-positivity is equivalent to nonnegativity of the Choi quadratic form on inverse vectorizations of rank at most $r$, Equation (22) is nonnegative.

Apply the result first with $r = s = \operatorname{rank} C$ to obtain Equation (21). For $s \leq r$,

$$\begin{aligned} r\|C\|_2^2 + \frac{1}{r}|\operatorname{Tr} C|^2 - \left(s\|C\|_2^2 + \frac{1}{s}|\operatorname{Tr} C|^2\right) \\ = (r-s)\left(\|C\|_2^2 - \frac{|\operatorname{Tr} C|^2}{rs}\right) \geq 0. \end{aligned}$$

The trace–rank inequality $|\operatorname{Tr} C|^2 \leq s\|C\|_2^2$ proves the last step. If $C \neq 0$ and $s < r$, both factors in the last display are strictly positive, so equality is impossible. □

**Proposition 5.2** (Optimal coefficients)**.** *Let $r \geq 1$. Suppose* $\dim U \geq r$ *and* $\dim V \geq 2$*, or conversely. In an inequality*

$$\|\operatorname{Tr}_U C\|_2^2 + \|\operatorname{Tr}_V C\|_2^2 \leq a\|C\|_2^2 + b|\operatorname{Tr} C|^2 \qquad (\operatorname{rank} C \leq r),$$

*one must have $a \geq r$. Once $a = r$ is fixed, one must have $b \geq 1/r$.*

*Proof.* Let $P_r$ be a rank-$r$ projection on $U$ and let $v, w \in V$ be unit vectors. For

$$C = P_r \otimes |v\rangle\langle w|$$

one has

$$\begin{aligned} \|\operatorname{Tr}_U C\|_2^2 &= r^2, & \|\operatorname{Tr}_V C\|_2^2 &= r|\langle w, v\rangle|^2, \\ \|C\|_2^2 &= r, & |\operatorname{Tr} C|^2 &= r^2|\langle w, v\rangle|^2. \end{aligned}$$

Taking $v \perp w$ forces $a \geq r$. With $a = r$, taking $v = w$ forces $b \geq 1/r$. The same matrices attain equality in Equation (20). □

5.1. **A complementary balanced-factorization lift.** Fraser, Huber, Pozsgay, and Vona prove Equation (20) directly from a balanced rank decomposition [6, Lem. 7.1 and §8]. Their proof exposes a second lifting principle. The following formulation is the abstraction of the two properties used in their argument.

**Proposition 5.3** (Balanced-polarization lift)**.** *Let $\mathcal{Y}$ be a finite-dimensional complex Hilbert space and let $\mathcal{L} : \mathcal{L}(H) \to \mathcal{Y}$ be linear. Suppose that, for all $x, y, u, v, s, t \in H$,*

$$\|\mathcal{L}(xy^*)\|^2 \leq \|x\|^2\|y\|^2 + |y^*x|^2, \tag{23}$$

$$\langle \mathcal{L}(uv^*), \mathcal{L}(st^*)\rangle = \langle \mathcal{L}(tv^*), \mathcal{L}(su^*)\rangle. \tag{24}$$

*If* $C \neq 0$ *and* $q = \operatorname{rank} C$*, then*

$$\|\mathcal{L}(C)\|^2 \leq q\|C\|_2^2 + \frac{1}{q}|\operatorname{Tr} C|^2. \tag{25}$$

*Consequently the right side may be replaced by* $r\|C\|_2^2 + |\operatorname{Tr} C|^2/r$ *whenever* $q \leq r$*.*

*Proof.* Choose a balanced factorization

$$C = \sum_{i=1}^{q} x_i y_i^*, \qquad X^*X = Y^*Y = G = (g_{ij}), \qquad y_i^* x_i = \tau := \frac{\operatorname{Tr} C}{q}.$$

It is obtained by splitting the singular values equally between $X$ and $Y$, then applying the same unitary to their columns so that the diagonal of $Y^*X$ is constant. The latter step is the Parker–Fillmore constant-diagonal theorem [5, 10]. Put $\delta_i = g_{ii} = \|x_i\|^2 = \|y_i\|^2$.

The diagonal terms in the expansion of $\|\sum_i \mathcal{L}(x_i y_i^*)\|^2$ are at most $\delta_i^2 + |\tau|^2$ by Equation (23). For $i \neq j$, Equation (24), Cauchy–Schwarz, and another application of Equation (23) give

$$|\langle \mathcal{L}(x_i y_i^*), \mathcal{L}(x_j y_j^*)\rangle| \leq \delta_i \delta_j + |g_{ij}|^2.$$

Therefore

$$\|\mathcal{L}(C)\|^2 \leq \left(\sum_i \delta_i\right)^2 + q|\tau|^2 + 2\sum_{i<j} |g_{ij}|^2.$$

On the other hand,

$$\|C\|_2^2 = \operatorname{Tr}(G^2) = \sum_i \delta_i^2 + 2\sum_{i<j} |g_{ij}|^2.$$

Subtracting the preceding upper bound from the right side of Equation (25) gives the explicit defect bound

$$q\|C\|_2^2 + \frac{1}{q}|\operatorname{Tr} C|^2 - \|\mathcal{L}(C)\|^2 \geq \sum_{i<j} (\delta_i - \delta_j)^2 + 2(q-1)\sum_{i<j} |g_{ij}|^2 \geq 0. \tag{26}$$

The extension from $q$ to $r$ is the same trace–rank calculation used after Equation (22). □

For the partial-trace application, take

$$\mathcal{L}(C) = (\operatorname{Tr}_U C, \operatorname{Tr}_V C)$$

in the Hilbert direct sum. Then Equation (23) is the rank-one partial-trace inequality; its defect is the expectation of $(I - F_U) \otimes (I - F_V) = 4P_U^- \otimes P_V^-$. Equation (24) is the four-state marginal identity. Thus this route and the Choi lift use the same double-antisymmetric geometry but extract different information from it.

The mechanisms are not reformulations of one another. The balanced lift requires the rigid four-linear symmetry Equation (24), but neither complete positivity nor a Choi norm; it gives a direct norm inequality. Theorem 1.1 requires a rank-two Choi bound and, for the protected term, even transpose parity, but no crossed identity; it returns an $r$-positive map. Both arguments nevertheless contain the same complete graph. In Equation (26) the first sum is the Dirichlet energy of $K_q$, while in Equation (12) its degree $q-1$ appears by grouping edge modes at vertices. The constant diagonal isolates the trace direction in the balanced proof; range orthogonality to $\delta_e$ protects the corresponding direction in the Choi proof.

There is also a common operator language. Let $\Gamma = \mathcal{L}^*\mathcal{L}$ on the Hilbert–Schmidt space and $\Omega = \sum_i e_i \otimes \bar{e}_i$. In matrix-unit coordinates, Equation (24) is exactly the reshuffling symmetry

$$\Gamma_{ij,kl} = \Gamma_{lj,ki}.$$

The rank-one hypothesis is block positivity of $I + |\Omega\rangle\langle\Omega| - \Gamma$ on product vectors; the conclusion is block positivity of $rI + |\Omega\rangle\langle\Omega|/r - \Gamma$ on vectors of Schmidt rank at most $r$. Thus the balanced argument is itself a low-rank positivity lift, based on a crossing-fixed positive Gram operator and a level-one input rather than a CP Choi operator and a level-two input.

**Corollary 5.4** (Spectral stability)**.** *Let $C \in \mathcal{L}(U \otimes V)$ have rank $q \geq 2$, let $\sigma_1, \ldots, \sigma_q$ be its nonzero singular values, and put $\overline{\sigma} = q^{-1} \sum_i \sigma_i$. If*

$$D_q(C) := q\|C\|_2^2 + \frac{1}{q}|\operatorname{Tr} C|^2 - \|\operatorname{Tr}_U C\|_2^2 - \|\operatorname{Tr}_V C\|_2^2,$$

*then*

$$\sum_{i=1}^{q} (\sigma_i - \overline{\sigma})^2 \leq \frac{D_q(C)}{q-1}. \tag{27}$$

*Proof.* In the balanced construction, $G$ is unitarily similar to $\operatorname{diag}(\sigma_1, \ldots, \sigma_q)$. If $\overline{\delta} = q^{-1} \sum_i \delta_i = \overline{\sigma}$, then

$$\|G - \overline{\delta} I\|_2^2 = \frac{1}{q} \sum_{i<j} (\delta_i - \delta_j)^2 + 2 \sum_{i<j} |g_{ij}|^2.$$

Multiplying by $q-1$ and comparing with Equation (26) proves Equation (27). ∎

## 6. A second instance: graph Kraus dictionaries

The next family does two jobs. It shows that the lifting theorem applies away from the partial-trace identity, and it proves that $(r-1)\beta_2$ is an attained universal coefficient rather than merely a proof artifact.

Let $G = (V, E)$ be a nonempty simple graph on $V = \{1, \ldots, m\}$. For $e = \{i, j\}$, $i < j$, put

$$L_e = E_{ij} - E_{ji} \in M_m, \qquad J = E_{01} - E_{10} \in M_2, \qquad A_e = L_e \otimes J,$$

and define

$$\Phi_G(X) = \sum_{e \in E} A_e X A_e^*. \tag{28}$$

Every $A_e$ is real symmetric. For $K_c = \sum_e c_e A_e = L_c \otimes J$, the nonzero singular values of the skew matrix $L_c$ occur in equal pairs. Since $\|L_c\|_2^2 = 2\|c\|_2^2$ and $J$ is unitary,

$$s_1(K_c)^2 + s_2(K_c)^2 = 2s_1(L_c)^2 \leq 2\|c\|_2^2.$$

A coefficient vector supported on one edge attains equality. Hence

$$\beta_2(\Phi_G) = 1. \tag{29}$$

For an integer $R \geq 1$, set

$$\Theta_{R,\lambda}^G = \Phi_G \circ \tau + \lambda \left( \Delta - \frac{1}{R} \operatorname{id} \right),$$

and set

$$\lambda_R^*(G) := \inf\{\lambda \geq 0 : \Theta_{R,\lambda}^G \text{ is } R\text{-positive}\}.$$

**Theorem 6.1** (Induced-density obstruction and clique sharpness)**.** *For every nonempty $S \subseteq V$ with $|S| \leq R$,*

$$\frac{2|E(G[S])|}{|S|} \leq \lambda_R^*(G) \leq R - 1. \tag{30}$$

*In particular, if $G$ contains an $R$-clique, then*

$$\lambda_R^*(G) = R - 1.$$

*Proof.* The upper bound is Theorem 1.1 and Equation (29). For the lower bound, take $C = P_S \otimes E_{01}$, where $P_S = \sum_{i \in S} E_{ii}$. Then

$$\operatorname{rank} C = |S|, \qquad \operatorname{Tr} C = 0, \qquad \|C\|_2^2 = |S|.$$

Because the $A_e$ are symmetric, the Choi convention gives

$$\langle \operatorname{vec} C, J(\Phi_G \circ \tau) \operatorname{vec} C \rangle = \sum_{e \in E} \langle A_e C, (A_e C)^T \rangle_{\mathrm{HS}}. \tag{31}$$

If both endpoints of $e$ lie in $S$, then

$$A_e C = -L_e \otimes E_{11}, \qquad (A_e C)^T = -A_e C,$$

so its contribution to Equation (31) is $-\|L_e\|_2^2 = -2$. If exactly one endpoint lies in $S$, then $L_e P_S$ is, up to sign, a single off-diagonal matrix unit; its Hilbert–Schmidt pairing with its transpose is zero. If neither endpoint lies in $S$, then $A_e C = 0$. The correction contributes $\lambda |S|$, and nonnegativity forces Equation (30). If $S$ is an $R$-clique, its induced average degree is $R - 1$, matching the universal upper bound. □

There is a useful warning in the most natural compression of this graph family. Let

$$\mathcal{S}_G = \operatorname{span}\{E_{ii}\} + \operatorname{span}\{E_{ij} : \{i,j\} \in E\},$$

and write $p = |0\rangle\langle 0|$, $q = |1\rangle\langle 1|$. Compressing input $X \otimes q$ to the $p$ output corner recovers the graph reduction map, but it annihilates the input itself. Under this compression,

$$\Delta_{2m} - \frac{1}{R}\mathrm{id} \quad \longmapsto \quad \Delta_m.$$

Thus the one-layer compression preserves the unprotected theorem Equation (4) and discards precisely the $-\mathrm{id}/R$ term. Keeping both diagonal qubit layers repairs the loss. The lesson is broader than this example: a coefficient-space reduction must preserve the protected vector, not only the Kraus action.

## 7. Sharp boundaries of the method

7.1. **Odd transpose parity.** Transposition grades the ambient matrix space through $\mathbb{C}^d \otimes \mathbb{C}^d = \operatorname{Sym}^2 \mathbb{C}^d \oplus \wedge^2 \mathbb{C}^d$. A Kraus space need not itself be transpose-invariant. The protected hypothesis selects the even summand: symmetric matrices have even degree, skew-symmetric matrices odd degree, and tensor-product degrees add modulo two. The cancellation in the proof of Equation (9) occurs exactly in even degree.

The even-parity hypothesis cannot be dropped from the universal protected conclusion. Let $H = \mathbb{C}^2$ and

$$A = E_{01} - E_{10}, \qquad \Phi(X) = AXA^*.$$

The one-dimensional Kraus space is skew. At $s = 2$, using the standard bases in Equation (6),

$$(A \otimes I)\eta_{12} = -\frac{1}{\sqrt{2}}\delta_e, \qquad (\Phi \otimes \mathrm{id})(P_-) = \Pi_e.$$

Consequently,

$$(\Phi \otimes \mathrm{id})(P_-) \preceq \lambda(I - \Pi_e)$$

fails for every finite $\lambda$: the right side vanishes on $\delta_e$ and the left side does not. The unprotected bound is sharp, since $\Pi_e \preceq I$ and $\beta_2(\Phi) = 1$.

7.2. **A Choi constant is not a contraction identity.** Theorem 1.1 returns an $r$-positive map. It does not say which concrete quadratic form that map represents. The partial traces in Equation (20) enter through Equation (19) and Equation (22), not through $\beta_2$. The zero map makes the logical separation immediate: it satisfies the theorem with $\beta_2 = 0$, but carries no information about either marginal. Any proposed new application therefore has two independent obligations: compute the local Choi constant and identify the downstream quadratic form.

7.3. **The universal coefficient need not be the exact coefficient.** The vertex Cauchy–Schwarz step Equation (12) discards nonnegative variance terms. It follows that $(r-1)\beta_2(\Phi)$ is a universal sufficient coefficient, not a formula for the optimum of every map. The clique case of Theorem 6.1 shows that no smaller universal coefficient is possible. For a particular Kraus dictionary without an appropriate witness, its exact correction threshold remains a separate extremal problem.

7.4. **Higher tensor powers.** An even tensor product of skew Kraus spaces still has even transpose parity, so Theorem 1.1 applies whenever its $\beta_2$ is known. This gives alternating marginal inequalities for higher even tensor powers. Two features of the two-factor application do not automatically persist:

(1) The constants $\beta_2(\Lambda_{U_1} \otimes \cdots \otimes \Lambda_{U_{2m}})$ are not determined by Theorem 4.1 when $m \geq 2$.
(2) The collapse in Equation (19) is special to two factors. If $S = \Delta - \mathrm{id}$ and one tries to match $S^{\otimes n} + (r-1)\Delta^{\otimes n} - t\mathrm{id}$ with a scalar multiple of $(S + t\mathrm{id})^{\otimes n}$, the terms containing $n-1$ copies of $S$ force $rt = r-1$, while those containing $n-2$ copies force $r - 1 = (r-1)^2/r$. For $n \geq 3$, these equations are compatible only at $r = 1$.

The lifting theorem therefore survives in higher even degree, while the particularly simple tensor-product reduction map does not.

## 8. Formal verification and scope

The finite-coordinate development proves Theorem 4.1, both the protected and unprotected parts of Theorem 1.1, Theorem 1.2, Theorems 5.1, 5.4 and 6.1, the exact-rank extension, both optimality witnesses, and the parity counterexample. The accompanying repository [11] is cited at the immutable revision inspected for this manuscript. It also proves the Autonne–Takagi factorization used in Section A. The main declarations contain no `sorry`; their reported axiom surface is `propext`, `Classical.choice`, and `Quot.sound`.

The formal statements use matrices over finite coordinate types. They prove the coordinate identities and rank bounds used in this paper, including $\mathrm{SR}(\operatorname{vec} C) = \operatorname{rank} C$. The transport from abstract finite-dimensional Hilbert spaces to chosen orthonormal coordinates is not a separate formal theorem. Accordingly, the kernel checks the deduction once those coordinate definitions are accepted; the correspondence between those definitions and the basis-free prose remains an interface to inspect.

For the balanced-polarization route in Section 5.1, the cited revision checks the generic lift, its complete-graph defect, the sharp rank-one marginal estimate, the crossed-polarization identity, and the unconditional partial-trace specialization. It also constructs the exact-rank equal-Gram factorization by spectral mixing and fourth-root rescaling, and checks that constant-diagonal unitary mixing balances it without changing the represented matrix. The trace-entry construction and Parker–Fillmore constant-diagonal induction are formalized as part of that route. Starting from the exact-rank singular factorization, the same unitary mixing preserves the centered Gram norm and identifies it with the variance of the positive singular values, yielding Theorem 5.4. The zero-map boundary and the incompatible higher-tensor coefficient equations in Section 7 are also packaged as guarded theorems.

## Appendix A. Proof of the double-skew action bound

We prove Theorem 4.1, adapting the argument of [7, §2] and reorganizing it around the Autonne–Takagi factorization. Relative to the fixed product basis, let $T_U$ and $T_V$ denote the two partial transposes on $\mathcal{L}(U \otimes V)$. They are commuting self-adjoint involutions for the Hilbert–Schmidt inner product, and

$$\Pi = \frac{1}{2}(1 - T_U)\frac{1}{2}(1 - T_V)$$

is the orthogonal projection onto $\mathfrak{so}(U) \otimes \mathfrak{so}(V)$. Hence $\Pi K = K$. Moreover $K^T = K$, because each elementary tensor contains two skew factors.

Autonne–Takagi factorization [8, Cor. 4.4.4(c)] gives

$$K = \sum_i s_i w_i w_i^T, \tag{32}$$

where the $w_i$ form an orthonormal basis, $s_i \geq 0$, and zero values of $s_i$ are retained. The matrices $w_i w_i^T$ are Hilbert–Schmidt orthonormal.

We first establish the two-term estimate. Write $M_u$ for the coefficient matrix of $u \in U \otimes V$. Direct contraction of the two product indices gives

$$\langle uu^T, (vv^T)^{T_U} \rangle_{\mathrm{HS}} = \operatorname{Tr}\big((M_u^* M_v)^2\big). \tag{33}$$

For orthonormal $w_1, w_2$, let

$$N = s_1 w_1 w_1^T + s_2 w_2 w_2^T, \qquad c_i = \|M_i^* M_i\|_2.$$

Expanding Equation (33),

$$\langle N, N^{T_U} \rangle_{\mathrm{HS}} = s_1^2 c_1^2 + 2 s_1 s_2 \operatorname{Re} \operatorname{Tr}\big((M_1^* M_2)^2\big) + s_2^2 c_2^2.$$

The Hilbert–Schmidt Cauchy–Schwarz inequalities

$$|\operatorname{Tr}(X^2)| \leq \|X\|_2^2, \qquad \|A^* B\|_2^2 = \langle AA^*, BB^* \rangle_{\mathrm{HS}} \leq \|A^* A\|_2 \|B^* B\|_2$$

bound the middle term from below by $-2 s_1 s_2 c_1 c_2$. Therefore

$$\langle N, N^{T_U} \rangle_{\mathrm{HS}} \geq (s_1 c_1 - s_2 c_2)^2 \geq 0. \tag{34}$$

Since $N^T = N$, one has $N^{T_U} = N^{T_V}$, and expansion of $\Pi$ gives

$$\langle N, \Pi N \rangle_{\mathrm{HS}} = \frac{1}{2}\left(\|N\|_2^2 - \langle N, N^{T_U} \rangle_{\mathrm{HS}}\right) \leq \frac{1}{2}\|N\|_2^2. \tag{35}$$

Now fix $a \neq b$ and truncate Equation (32) to

$$K_{ab} = s_a w_a w_a^T + s_b w_b w_b^T, \qquad A = s_a^2 + s_b^2.$$

Orthogonality gives $\|K_{ab}\|_2^2 = \langle K, K_{ab} \rangle_{\mathrm{HS}} = A$. Since $\Pi$ is a self-adjoint projection and $\Pi K = K$,

$$A = \langle K, \Pi K_{ab} \rangle_{\mathrm{HS}} \leq \|K\|_2 \|\Pi K_{ab}\|_2.$$

By Equation (35), $\|\Pi K_{ab}\|_2^2 = \langle K_{ab}, \Pi K_{ab} \rangle_{\mathrm{HS}} \leq A/2$. Thus $A^2 \leq A\|K\|_2^2/2$, and cancellation when $A > 0$ yields

$$s_a^2 + s_b^2 \leq \frac{1}{2}\|K\|_2^2 \qquad (a \neq b). \tag{36}$$

Finally, Equation (32) implies $K^* K = \sum_i s_i^2 |\overline{w}_i\rangle\langle\overline{w}_i|$. For orthonormal $x, y$, set

$$\alpha_i = |\langle \overline{w}_i, x \rangle|^2 + |\langle \overline{w}_i, y \rangle|^2.$$

Then $0 \leq \alpha_i \leq 1$ and $\sum_i \alpha_i = 2$. The polytope

$$\{\alpha : 0 \leq \alpha_i \leq 1,\ \sum_i \alpha_i = 2\}$$

is the convex hull of the vectors having two coordinates equal to one and the rest zero. Hence Equation (36) gives

$$\|Kx\|^2 + \|Ky\|^2 = \sum_i \alpha_i s_i^2 \le \max_{a\ne b}(s_a^2 + s_b^2) \le \frac{1}{2}\|K\|_2^2.$$

This proves Theorem 4.1.

## Appendix B. Elementary Choi contractions

For completeness, we record the four contractions used in Equation (22). With the vectorization convention of Section 3,

$$\begin{aligned}
\langle \operatorname{vec} C, J(\Delta_U \otimes \Delta_V) \operatorname{vec} C\rangle &= \|C\|_2^2,\\
\langle \operatorname{vec} C, J(\mathrm{id}_U \otimes \Delta_V) \operatorname{vec} C\rangle &= \|\mathrm{Tr}_U\, C\|_2^2,\\
\langle \operatorname{vec} C, J(\Delta_U \otimes \mathrm{id}_V) \operatorname{vec} C\rangle &= \|\mathrm{Tr}_V\, C\|_2^2,\\
\langle \operatorname{vec} C, J(\mathrm{id}_U \otimes \mathrm{id}_V) \operatorname{vec} C\rangle &= |\mathrm{Tr}\, C|^2.
\end{aligned}$$

They follow by expanding $C$ in matrix units and matching indices. Their order depends only on the convention that $\mathrm{Tr}_U$ traces out $U$; interchanging $U, V$ interchanges the middle two lines and leaves every result unchanged.

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